CASE STUDY REPORT

Project Title:
Home Service Booking and Management System

Project Name:
“K-Handyman”

Prepared By:
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Project Type:
Case Study of the Project Made By Me

ABSTRACT

This project presents the design and development of “Handyman in Kathmandu,” a modern web-based service platform developed using the MERN Stack — MongoDB, Express.js, React.js, and Node.js. The platform aims to connect users in Kathmandu with skilled service providers offering a variety of home maintenance and repair services, such as plumbing, electrical work, cleaning, carpentry, painting, and appliance repair.

Focusing on usability, scalability, and responsiveness, the frontend is built with React.js to deliver a dynamic and intuitive user interface. It enables customers to easily browse available services, view provider details, and book appointments in real time. The backend, developed with Node.js and Express.js, handles authentication, booking logic, and communication between clients and service providers, while MongoDB efficiently manages and stores all service, user, and booking data.

Core features of the platform include user authentication (admin, provider, and customer roles), service management, real-time booking, and secure data handling. The site also implements responsive design principles to ensure accessibility across devices, from desktops to mobile phones.

By integrating both front-end and back-end technologies, “Handyman in Kathmandu” demonstrates how a full-stack JavaScript framework can be leveraged to build a robust, interactive, and efficient service platform — bridging the gap between customers seeking reliable help and professionals offering trusted services.

Keywords: ***Electronic Product, Customer Satisfaction, Customer Loyalty, E-commerce Experience***

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CSS	Cascading Style Sheets
DBMS	Database Management System
ETA	Estimated Time of Arrival
FCFS	First Come, First Served
FCM	Firebase Cloud Messaging
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
JS	JavaScript
JSON	JavaScript Object Notation
JWT	JSON Web Token
KNN	K-Nearest Neighbors
MERN	MongoDB, Express.js, React.js, Node.js
ODM	Object Data Modeling
REST	Representational State Transfer
UI	User Interface

CHAPTER 1: INTRODUCTION

1.1 Project Introduction

The rapid growth of online service platforms has transformed how people access local professionals for home maintenance and repairs, enabling service providers to reach wider audiences and customers to find reliable help quickly. In cities like Kathmandu—where convenience, trust, and skilled labor are essential—an efficient and user-friendly digital platform is vital to connect clients with qualified handymen. This project, titled "Handyman in Kathmandu," focuses on building a modern, responsive, and full-stack web application designed to meet the everyday service needs of urban households.

The platform is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), chosen for its flexibility, scalability, and seamless integration between the client and server. These technologies enable the creation of a visually appealing interface, dynamic features, and secure backend operations. Users can browse available services, view provider details, make bookings, and manage appointments with ease.

Handyman in Kathmandu offers a range of professional services, including plumbing, electrical work, cleaning, carpentry, painting, and appliance repair. Each service category is clearly organized to help users find what they need quickly and efficiently. Key features include real-time service booking, user authentication, search and filtering, and database-driven management for both customers and providers.

This project combines essential service management functionality with a clean and responsive interface, demonstrating how a full-stack JavaScript framework can power a complete digital service solution. It establishes a strong foundation for future upgrades such as online payments, real-time chat, and automated service recommendations to further enhance user experience and system efficiency.

1.2 Problem Statement

In Kathmandu, many individuals face difficulties finding reliable and skilled handymen for essential home maintenance and repair tasks. The traditional approach of searching through personal networks or local advertisements often leads to uncertainty, time delays, and lack of trust between customers and service providers. Additionally, there is no centralized digital platform that allows users to compare services, check provider availability, and book appointments seamlessly.

From the service provider's perspective, many skilled workers struggle with limited online visibility and inefficient job management, preventing them from reaching potential customers and organizing their bookings effectively.

This project addresses these challenges by developing a web-based platform using the MERN stack that connects customers with verified handymen, ensures transparency in service selection, and simplifies the overall booking and management process. The system's relevance lies in its ability to digitize and streamline the home service industry, improving convenience, trust, and efficiency for both customers and service providers in Kathmandu.

1.3 Objective

The overall objective of the project is to:

- To build an easy-to-use MERN web platform that connects customers with verified handymen in Kathmandu.
- To provide a review-based rating system so customers can rate workers and help top-rated handymen appear first.
- To allow handymen to upload certificates for building trust and proving their skills.
- To create a simple, responsive interface for booking services, managing profiles, and giving feedback.
- To enable the admin to manage users, providers, bookings, and earnings through a dashboard with commission tracking.

1.4 Scope and Limitation

Scopes:

- The project focuses on developing a web-based platform that connects customers with

skilled handymen in Kathmandu for services such as plumbing, electrical work, cleaning, and carpentry.

- It is built using the MERN stack (MongoDB, Express.js, React.js, and Node.js) to ensure a smooth and scalable system.
- The platform allows users to register, book services, and give reviews, while service providers can manage their profiles and service requests.
- A review-based ranking system is included so that workers with higher ratings appear at the top of the search list, encouraging better service quality.
- The system is designed to be responsive and easy to use, providing access from both desktop and mobile browsers.

Limitations:

- Since the platform is new, it may take time to gain enough users and reviews for accurate rankings.
- There is no dedicated mobile application; users can only access the system through a web browser.
- Due to limited time and resources, features like push notifications, and automated scheduling were not implemented.
- The current version does not include an online payment gateway; all bookings are confirmed manually.

1.5 Development Methodology

System Development Model

The system development follows an Agile software development methodology, specifically using Scrum practices. This approach allows for incremental progress, quick feature delivery, and flexibility in adapting to user feedback and changing requirements during the development of the Handyman in Kathmandu platform.

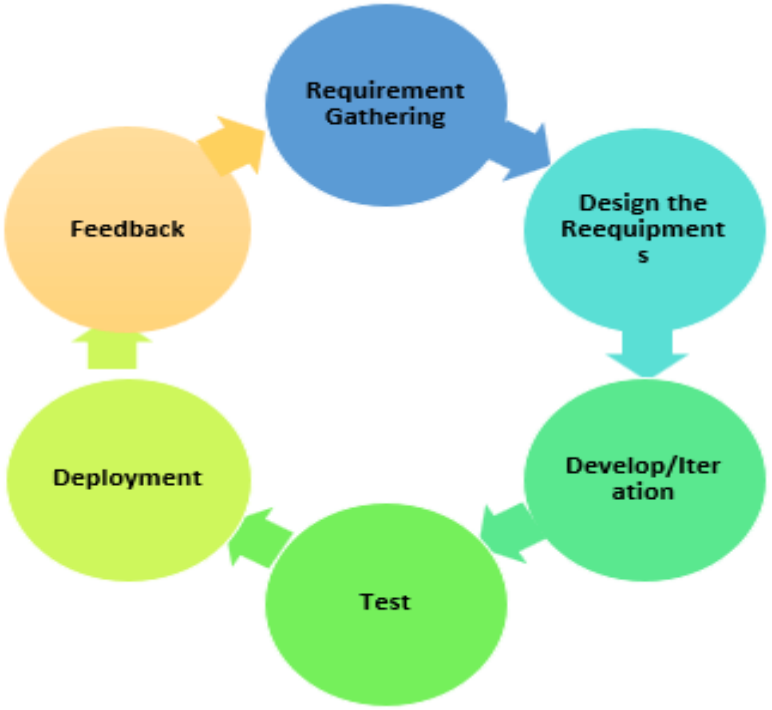


Figure:1 Agile Methodology Diagram

1.6 Report Organization

This report is organized into six chapters which give information about different related topics of the project.

Chapter 1:

Provides an introduction to the K-Handyman system. It explains the background of the problem, objectives, scope, and limitations of the project. It also includes an overview of the platform, its purpose, and the roles of each team member involved in the development process.

Chapter 2:

Focuses on the background study and literature review. It examines existing service-booking platforms, related research, and technologies that influenced the design and implementation of the K-Handyman system.

Chapter 3:

Covers the system analysis and requirement specifications. This includes the identification of functional and non-functional requirements, user role analysis, and a feasibility study covering technical, operational, and economic aspects.

Chapter 4:

Discusses the system design and architecture of the project. It includes UML diagrams, flowcharts, database design, user interface design, and explanations of key algorithms used—such as the review-based ranking system and priority-based service management.

Chapter 5:

Describes the implementation and testing phase. It explains how the frontend and backend were developed using the MERN stack, integration between components, and results of both unit and system testing. Screenshots, workflows, and test outcomes are also presented in this section.

Chapter 6:

Provides the conclusion and future scope of the project. It summarizes the outcomes, highlights major achievements, and suggests possible future improvements such as online payment integration, mobile app development, and AI-based service recommendations.

CHAPTER 2: BACKGROUND AND LITERATURE REVIEW

2.1 Background Study

In many parts of Nepal—especially in cities like Kathmandu, Pokhara, and Biratnagar—people still face difficulty finding reliable handymen for everyday tasks such as plumbing, electrical repairs, painting, and cleaning. Most depend on personal contacts or word-of-mouth recommendations, which are often inconsistent, time-consuming, and unreliable. This leads to delays, miscommunication, and inconvenience for both customers and service providers.

At the same time, many skilled handymen struggle to promote their services or manage bookings efficiently. Without an online presence, their visibility remains low, and they miss opportunities to reach new customers or build trust through verified reviews. This gap clearly shows the need for a structured digital platform that connects customers directly with trustworthy workers.

With the increasing use of smartphones and the growth of digital platforms in Nepal, online service-based systems are becoming more practical and widely accepted. In response to this need, K-Handyman was developed as a MERN-based web platform designed to help users easily find, book, and review trained handymen.

The system includes features such as user and handyman registration, booking management, service listings, and a review-based ranking system that highlights top-rated workers. It also allows handymen to upload training certificates, helping customers verify their authenticity. Additionally, fixed service pricing makes the process more transparent, and the admin earns a small commission from bookings while monitoring all activities through a dedicated dashboard.

By digitally linking customers with trained and verified handymen, K-Handyman improves access to reliable home services, increases trust and transparency, and provides workers with better exposure and consistent job opportunities. The platform supports everyday convenience while helping skilled professionals grow in Nepal’s rapidly developing digital economy.

2.2 Literature Review

In recent years, the use of digital platforms for on-demand household services has grown rapidly around the world. Applications that connect users with nearby service providers now rely on location-based search, real-time communication, and online booking systems to deliver fast and efficient services. Several studies and existing systems provide useful insights into how such platforms operate and the problems they aim to solve.

Shrestha and Adhikari [1] emphasized the increasing importance of digital handyman solutions in Nepal, driven by the challenges of urban growth and the lack of reliable service access. Their study pointed out that depending on informal recommendations often leads to delays, inconsistency, and a lack of trust, highlighting the need for a centralized, digital service-finding platform.

Rao et al. [2] explored the concept of location-based service matching, proposing a system that uses GPS data to identify and recommend nearby service providers. Their work showed that integrating map visualization and geolocation algorithms enhances accessibility and helps users make better decisions. They also demonstrated that methods like the Haversine formula and Backtracking can accurately calculate distances and rank providers in real time.

The importance of real-time communication in service applications has also been highlighted in several studies. Kumar and Singh [3] discussed the use of socket-based communication to handle live chat, booking updates, and instant notifications between users and service professionals. Their findings suggest that immediate feedback and interaction significantly improve user engagement and transparency.

To ensure fairness in handling multiple customer requests, Pandey et al. [4] presented a queue-based service model that assigns jobs using a First-Come, First-Served (FCFS) system. This approach helps balance workloads among workers and minimizes delays in service delivery.

Another essential component of service applications is the ratings and reviews system. According to Lee [5], user-generated feedback builds credibility and assists new customers in choosing reliable service providers. Incorporating a review-based ranking model was shown to increase trust and user satisfaction in similar platforms.

The role of cloud-based storage and digital identity verification has also been examined. Thapa et al. [6] found that using cloud storage for managing profile data and documents improves system scalability, security, and data retrieval efficiency, which are crucial for modern web-based systems.

From these studies, it is clear that a successful service platform combines location tracking, queue-based task management, real-time communication, and user review systems. The design of K-Handyman draws from these ideas, integrating review-based worker ranking, fair booking management, and responsive communication to create a reliable and efficient digital service platform tailored for Nepal's growing urban environment.

Chapter 3: SYSTEM ANALYSIS

3.1System Analysis

3.1.1 Requirement Analysis

Functional Requirements:

- **Role-Based Login and Registration:**
  The system provides separate access for customers and handymen. Each role has its own dashboard and permissions for performing actions.
- **Handyman Profile Management:**
  Handymen can create and manage their profiles by adding service types, working hours, and service areas to help customers find the right match.
- **Service Search and Discovery:**
  Customers can browse or search for handymen based on their service category, location, or availability.
- A**ppointment Booking and Request Form:**
  Users can book services by filling out a simple form describing their issue, preferred time, and service location.
- **Scheduling and Availability Tracking:**
  The system ensures no overlapping appointments using the backtracking algorithm and allows customers to view available time slots.
- **Real-Time Notifications:**
  Both customers and handymen receive instant alerts when an appointment is created, accepted, or cancelled, through Socket.IO.
- **Ratings and Reviews:**
  After completing a service, customers can rate and review handymen. These ratings are used in the Bayesian ranking algorithm to ensure fair visibility.
- **Dashboard for Both Roles:**
  A dedicated dashboard helps customers track their bookings and reviews, while handymen can view pending, accepted, and completed jobs efficiently.

3.1.2 Use Case Diagram

It shows the main users of the system and the actions they can perform, such as booking services, managing profiles, messaging, and giving reviews. It provides a clear overview of how customers, handymen, and admin interact with the platform.ainable solution for connecting customers with local handymen in Kathmandu.

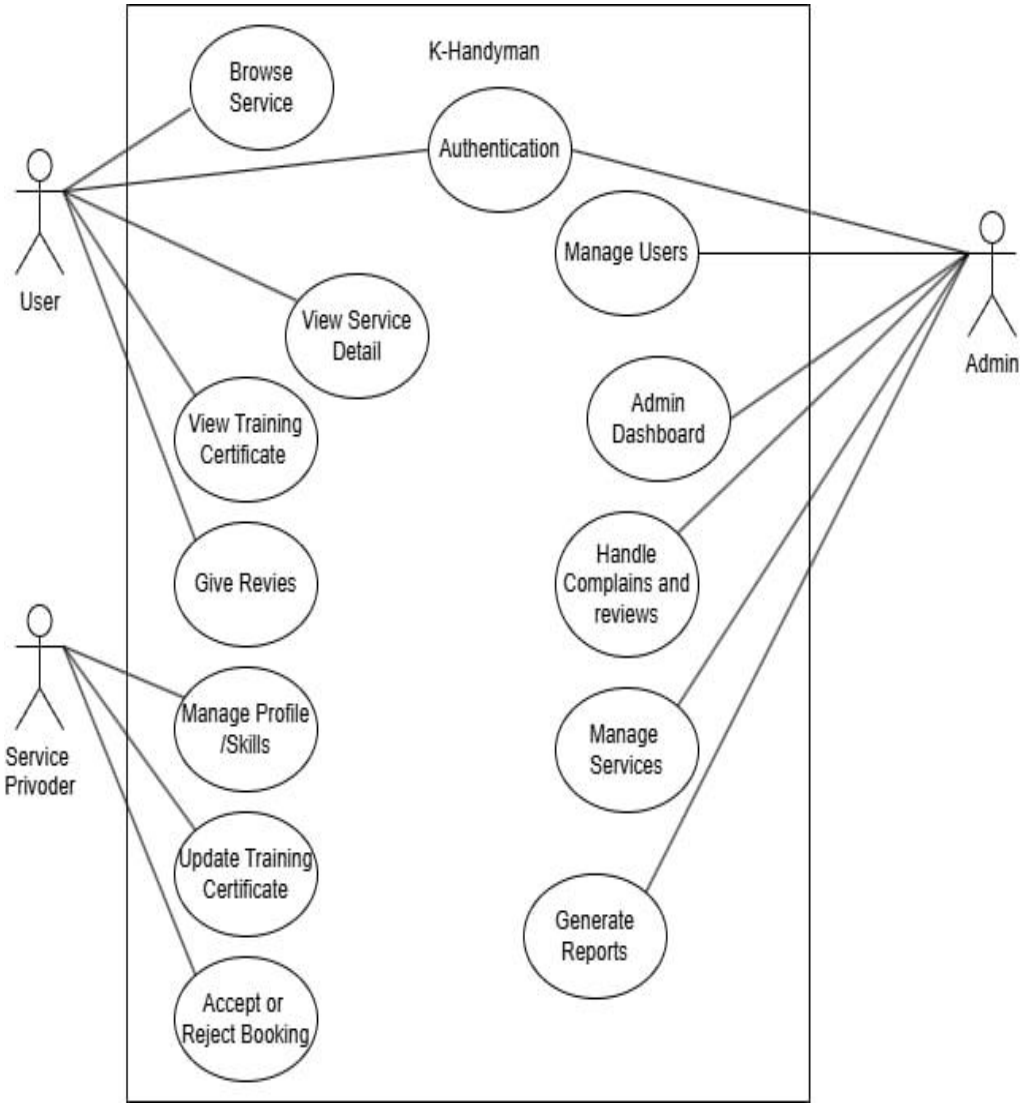


Figure:2 Use case diagram of K-Handyman

Non-Functional Requirements:

- Data Security: Implements JWT-based authentication and encrypted connections to ensure user information and booking details remain private and protected.
- Cross-Platform Compatibility: Designed to work smoothly across various browsers and devices, ensuring consistent performance on desktops, tablets, and smartphones.
- System Scalability: The backend can efficiently handle an increasing number of users, bookings, and service categories as the platform grows.
- Real-Time Responsiveness: Provides instant updates on booking confirmations, service status changes, and reviews to keep both users and handymen informed.
- User-Friendly Interface: Features a simple, intuitive layout that minimizes the learning curve and allows users to navigate and book services easily.
- High Availability: The system ensures reliable uptime and quick response times, maintaining smooth operation even during high traffic periods.

3.2 Feasibility Study

Some of the feasibility studies are as follows:

i. Technical Feasibility

For the K-Handyman system, the frontend is developed using React.js, along with HTML, CSS, and JavaScript, to create a responsive and user-friendly interface where customers can browse services, view handyman profiles, and make bookings easily. The backend is built using Node.js and Express.js, which handle server-side operations such as authentication, booking management, and API communication. Data is stored securely in MongoDB, ensuring efficient handling of user details, service records, and reviews.

ii. Operational Feasibility

The K-Handyman system operates as a web-based platform designed to connect customers with skilled handymen for various household services such as plumbing, electrical repair, cleaning, and carpentry. Currently, the system is hosted on a local server environment for development and testing purposes, which limits its accessibility to local users only. In future stages, the project aims to be deployed on a cloud-based hosting service with a

registered domain name, ensuring broader accessibility, improved performance, and enhanced data security..

iii. Economic Feasibility

The development of the K-Handyman system is economically feasible, as all major technologies used in the project—MongoDB, Express.js, React.js, and Node.js—are open-source and freely available online. This significantly reduces software licensing and development costs. Additionally, modern web hosting and domain registration services are affordable, making deployment and maintenance cost-effective. The project requires minimal financial investment while offering a scalable and sustainable solution for connecting customers with local handymen in Kathmandu.

iv. Schedule Feasibility

The development of the K-Handyman system is planned to be completed within the allocated time frame. The project timeline is considered realistic and achievable, as the development process is divided into manageable phases covering frontend design, backend implementation, database integration, and testing. Each module—such as user authentication, service booking, review system, and ranking algorithm—will be completed according to the predefined schedule. With proper coordination and adherence to the project plan, the system is expected to be delivered successfully within the estimated duration.

Phase	Task	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11-12
Planning & Design	Review Requirements & Scope						
Planning & Design	Finalize Database Schema						
Planning & Design	Design UI/X Mockups						
Backend Development	Setup Server & MONGODB						
Backend Development	User Authentication API						
Backend Development	Booking/Scheduling API						
Frontend Development	User Auth UI						
Frontend Development	Handyman Listings UI						
Frontend Development	Admin/Dashboard UI						
Final Prep & Submission	Documentation & Final Prep						
Final Prep & Submission	Bug Fixes & Iteration						
Final Prep & Submission	Final Documentation & Prep						
Final Prep & Submission	Deployment						
Final Prep & Submission	Project Submission						

Figure:3 Gantt Chart

Chapter 4: SYSTEM DESIGN

4.1 High-Level Design of System

The K-Handyman system is developed using a MERN stack (MongoDB, Express.js, React.js, Node.js) within a client–server architecture, providing a flexible and scalable environment for modern web applications. The project follows the Agile methodology to support continuous development, collaboration, and adaptability throughout the project lifecycle.

- **Incremental Progress:** The system is built gradually, with each iteration delivering functional modules such as user login, booking management, and review-based ranking.
- **Adaptive Planning:** The team adjusts goals and timelines based on feedback and changing requirements to ensure an efficient workflow.
- **Collaborative Development:** Regular communication between frontend and backend developers ensures smooth integration and consistent performance.
- **Task Prioritization:** Features and fixes are organized according to their importance and impact on user experience.
- **Quality Assurance:** Frequent testing and code reviews are conducted after everydevelopment phase to maintain system reliability and stability.

This Agile approach helps the K-Handyman team maintain flexibility, ensure quality, and deliver a system that effectively meets both user and business needs.

System Architecture:

- **Architecture Type:** The K-Handyman platform follows a client–server architecture developed as a web-based application to connect customers with service providers efficiently.
- **Frontend:** Built using React.js with CSS and JavaScript, ensuring an interactive, responsive, and user-friendly interface for both customers and handymen.
- **Backend:** Developed with Node.js and Express.js, managing business logic, API

routing, authentication, and communication between the client and database.

- **Database:** MongoDB Atlas is used for storing user profiles, service details, booking information, and customer reviews in a secure and scalable manner.
- **Real-Time Updates:** Implemented using WebSocket (Socket.io) to enable instant booking updates and service status notifications.
- **Review-Based Ranking:** The system ranks handymen dynamically based on customer ratings, ensuring that top-performing workers appear first in search results.
- **Deployment:** The application is deployed on Vercel for the frontend and Render for the backend, providing reliable hosting, scalability, and easy maintenance.

4.1.1 Flowchart

It shows the step-by-step process of how a user books a handyman — starting from login, selecting a service, sending a request, handyman approval, service completion, payment, and giving a review. It helps to clearly show the workflow of the K-Handyman system.

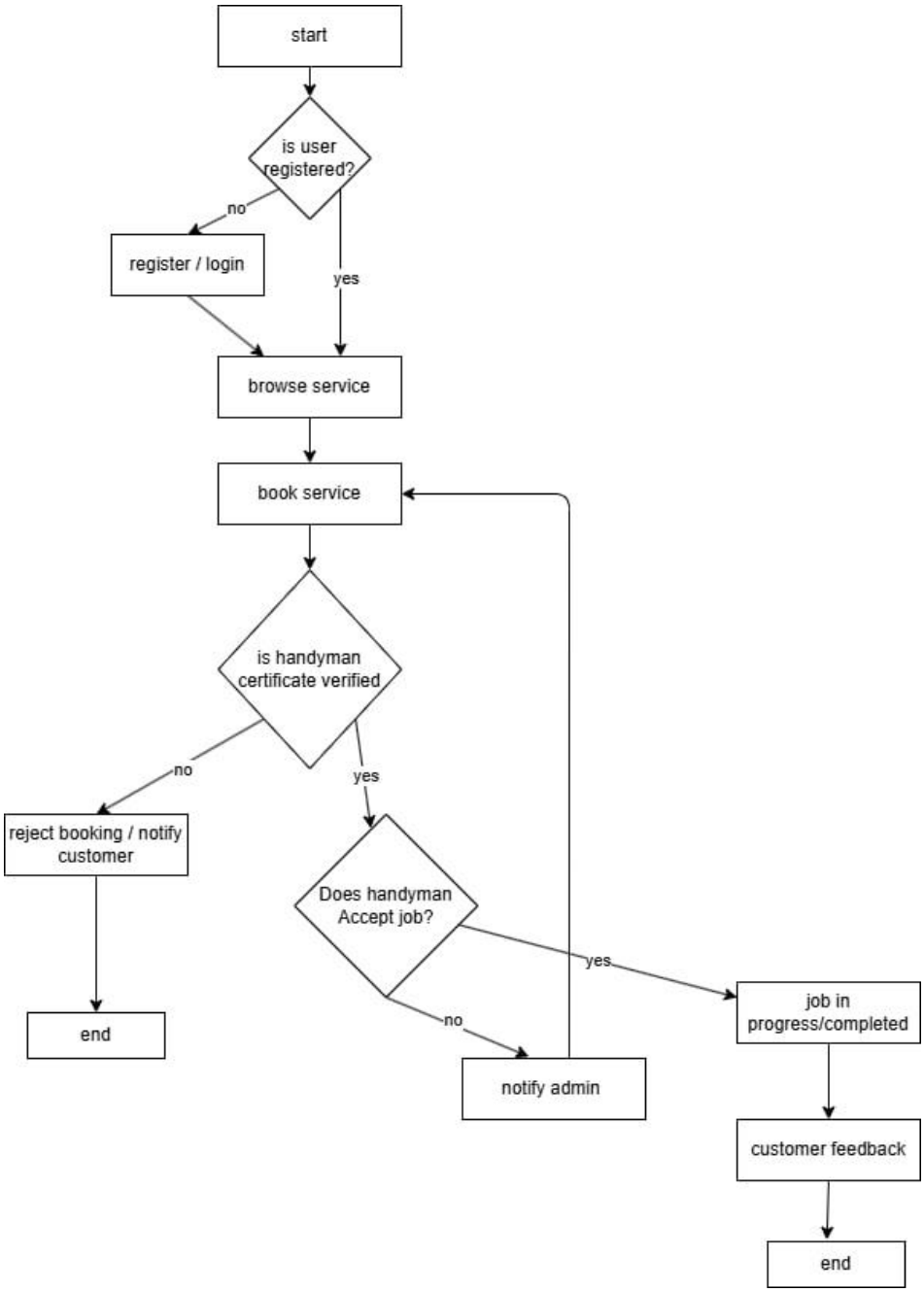


Figure:4 Flowchart of K-Handyman

4.1.2 Class Diagram

The class diagram of the K-Handyman system presents the main structural components involved in the platform and how they are connected. The system consists of several core classes: User, Handyman, Service, Booking, Message, and Review.

The User and Handyman classes store information related to account creation, login, and profile management. The Service class contains details about the different services offered by handymen, including service type, description, and pricing. The Booking class manages all activities related to service requests, scheduling, confirmations, and tracking the status of appointments.

Communication between customers and handymen is supported by the Message class, which handles chat-related data to enable smooth interaction during the service process. The Review class stores user feedback, ratings, and comments provided after a service is completed. These reviews play a key role in the system's ranking mechanism, helping promote reliable and highly rated handymen to customers.

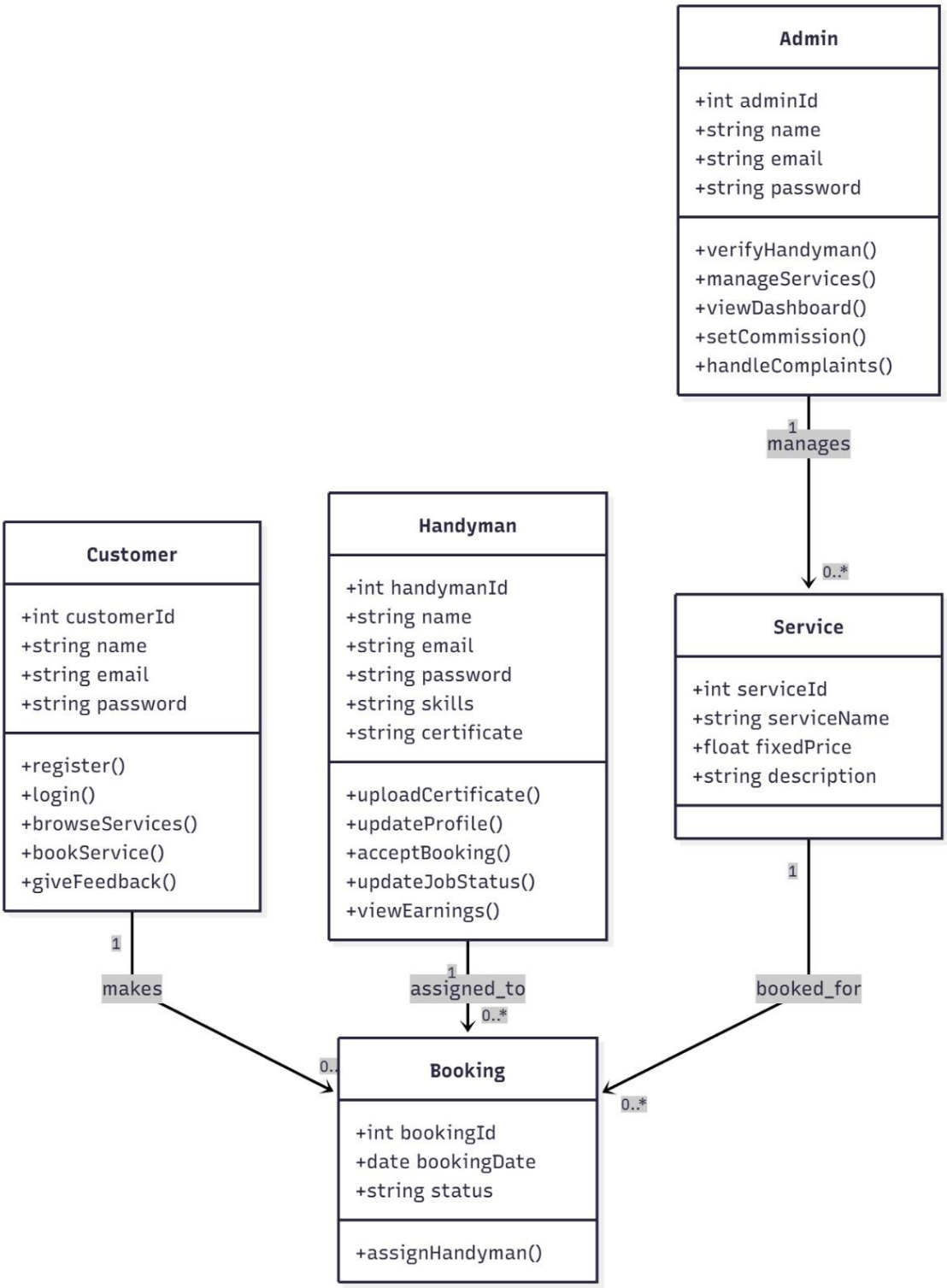


Figure:5 Class Diagram of K-Handyman

4.1.3 Sequence Diagram

A sequence diagram is an interaction diagram that shows how different objects in the system communicate with each other over time. It illustrates the sequence of messages and the order in which actions occur, representing the dynamic behavior of the system.

In the K-Handyman system, the sequence diagram demonstrates how a user searches for available handymen, selects a suitable service provider, creates a booking request, and how the handyman responds to that request. It also represents how booking confirmations, status updates, and reviews are exchanged between the user and handyman, ensuring smooth and transparent interaction throughout the service process.

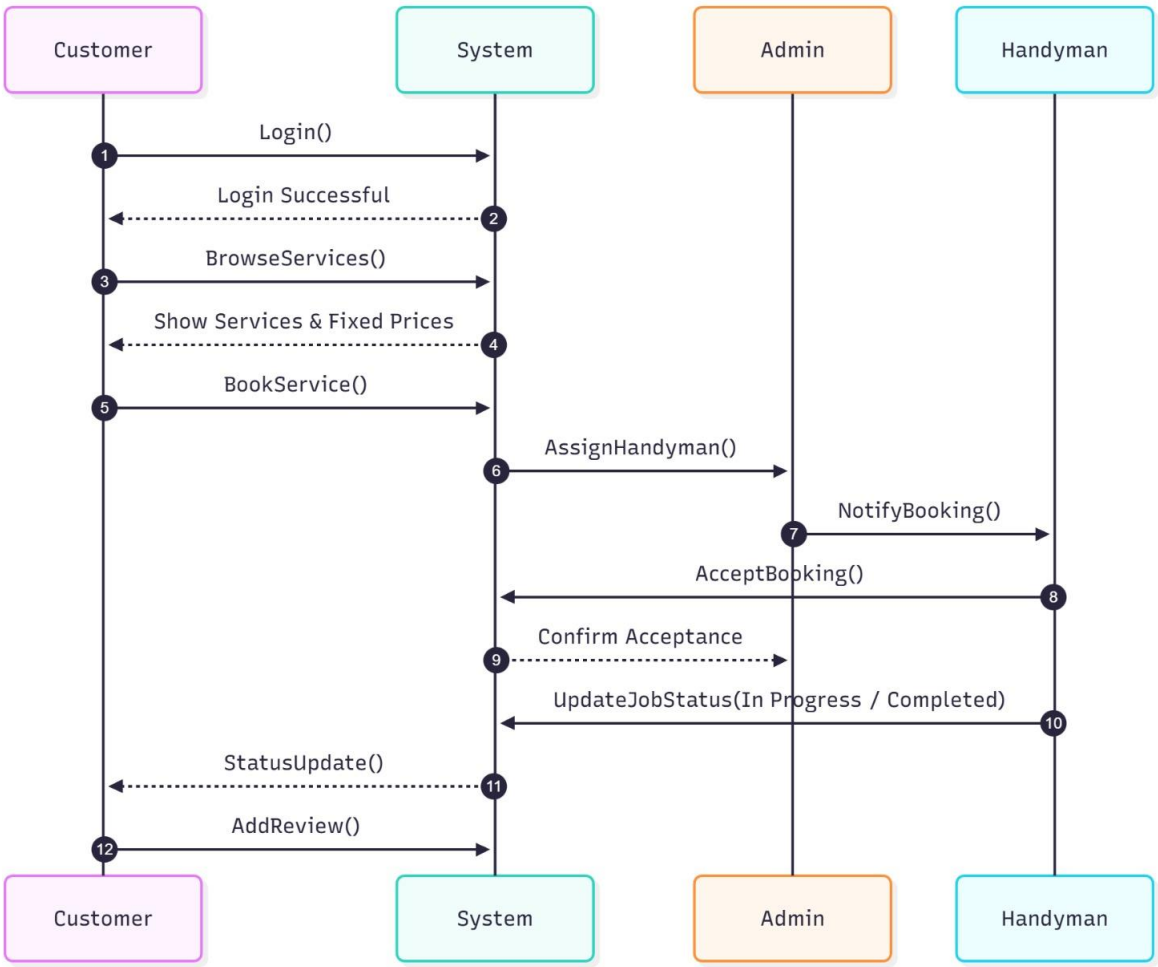


Figure:6 Sequence Diagram of K-Handyman

4.1.4 Activity Diagram

An Activity Diagram represents the dynamic behavior of the K-Handyman system. It provides a graphical view of the workflow, showing the sequence of actions and decisions involved in different processes. The diagram illustrates how control flows from one activity to another, such as user registration, service search, booking creation, handyman confirmation, and review submission. It helps visualize the overall process of how users and handymen interact within the platform from start to completion of a service.

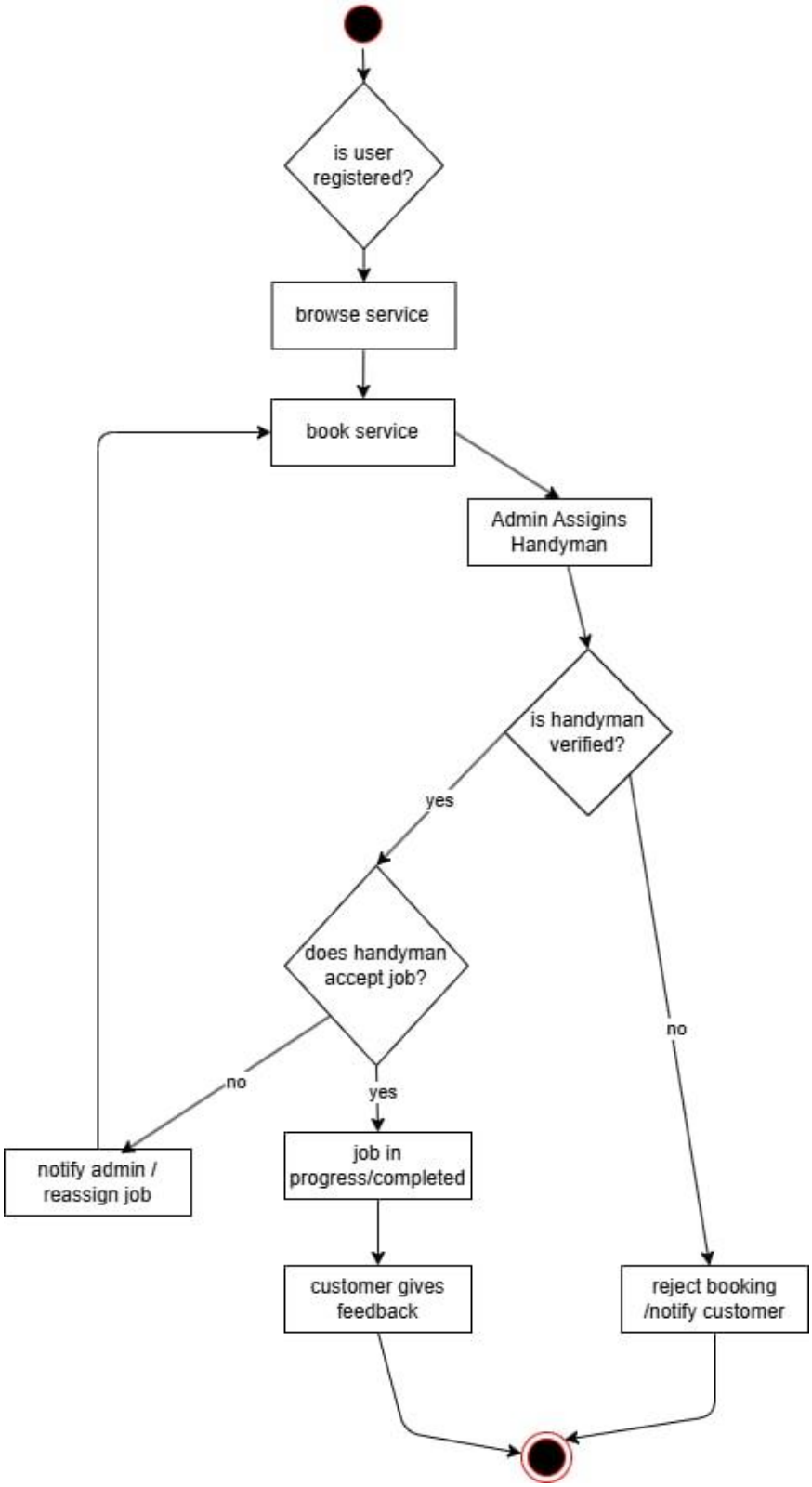


Figure:7 Activity Diagram of K-Handyman

4.2 Algorithms

The project uses two core algorithms to ensure fair ranking and reliable appointment scheduling: Bayesian rating and ranking model to highlight trustworthy handymen, and backtracking search to find feasible appointment slots under various service constraints.

4.2.1 Bayesian Rating & Ranking

To maintain fairness and avoid bias toward workers with few reviews, K-Handyman ranks providers using a Bayesian estimation model. This approach provides a statistically balanced ranking that considers both the average rating and the total number of reviews.

Binary mapping of ratings: Ratings of 4–5 stars are treated as positive, while 1–3 stars are treated as negative.

Let a handyman have *s* positive and *f* negative reviews.

Posterior (Beta–Bernoulli):

$$\text{Prior: Beta } (\alpha_0, \beta_0) \Rightarrow \text{Posterior Beta } (\alpha_0 + s,\ \beta_0 + f)$$

Where:

- α_0, β_0 = prior values (e.g., $\alpha_0 = \beta_0 = 1$)
- s = number of positive (4–5 star) reviews
- f = number of negative (1–3 star) reviews

This formula ensures reliable ranking results even when some handymen have few ratings.

4.2.2 Backtracking-Based Appointment Scheduling

When a customer requests a service, the system uses a Backtracking algorithm to find the most suitable handyman and available appointment slot that meets all constraints (availability, skill type, distance, and schedule).

Request model:

$$R = \left(svc,\ [t_{\min}, t_{\max}],\ L,\ d\right)$$

Candidate set: H* = all handymen who offer the requested service, are within range, and have available time slots.

Variables & constraints:

- **Decision variable:** assignment A= (h,t_{start}).
- **Constraints:**
  1. **Availability:** $\mathbf{t_{start}}$ in handyman's free slots, length d.
  2. **No overlap:** assigned interval [t_{start}, t_{start} + d] doesn't intersect existing bookings.
  3. **Caapacity:** daily jobs < limit.
  4. **Window :** $t_{min} < t_{start} < t_{max}$ - d.
  5. **Skill match:** handyman covers svc.

Search order (heuristics):

- Order H*H^*H* by scoreBayes\text{score}_{\text{Bayes}} scoreBayes(higher first).
- For each handyman, iterate candidate start times (sorted earliest-first).

- $d(V_i)$= Distance of vendor $V_i$from the user
- K= Number of nearest vendors to recommend (e.g., 3 or 5)

This ensures that users are presented with the most accessible vendors first.

4.2.3 Haversine Formula

The Haversine Formula is used in K-Handyman to calculate the distance between a customer and a handyman based on their geographic coordinates (latitude and longitude). This helps the system show the nearest and most relevant handymen to the user.

$$d = 2R \cdot \arcsin(\sqrt{\sin^2(\frac{\Delta\phi}{2}) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\frac{\Delta\lambda}{2})})$$

Where:

- R= Radius of Earth (≈ 6371 km)
- d = Distance between user and handyman
- ϕ_1, ϕ_2= Latitudes of user and vendor (in radians)
- λ_1, λ_2= Longitudes of user and vendor (in radians)
- $\Delta\phi = \phi_2 - \phi_1$
- $\Delta\lambda = \lambda_2 - \lambda_1$

By using the Haversine Formula, K-Handyman ensures users always see the nearest, fastest-reachable, and most convenient handyman options.

Chapter 5: IMPLEMENTATION AND TESTING

5.1 Implementation

The K-Handyman system is developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. This full-stack JavaScript framework offers a modern, scalable, and efficient environment for building responsive web applications. The project utilizes JavaScript as the main programming language, ensuring smooth communication between the frontend and backend components. Development and testing are carried out using Visual Studio Code, providing an integrated and user-friendly workspace for implementing, debugging, and managing the system's features.

5.1.1 Front-End Tools

The user interface of K-Handyman is crafted to deliver a smooth and interactive experience for both customers and service providers. The frontend is built using React.js, which allows the creation of reusable components and supports dynamic content rendering for faster page interactions. The layout is structured with HTML and styled using CSS, providing a clean, modern look that adapts well to different screen sizes and devices.

To improve functionality and maintainability, several supporting technologies are integrated into the frontend:

- React functional components and Hooks simplify state updates and data flow within the interface.
- React Router manages navigation, enabling users to switch between pages without reloading.
- Context API handles global states such as authentication, user roles, and notifications.
- Axios is implemented for sending and receiving data securely between the client and server.

Together, these tools help the application run efficiently in any web browser, offering users a responsive design, fast interactions, and a seamless experience when browsing, booking, or reviewing handyman services.

5.1.2 Back-End Tools

The backend of the K-Handyman system is developed using Node.js and Express.js, which together manage all server-side operations including API routing, authentication, and data handling. The backend ensures smooth communication between the user interface and the database while maintaining data security and system performance.

For data management, MongoDB is used as the primary database, providing a flexible and scalable structure to store information related to users, handymen, services, bookings, and reviews. The Mongoose library serves as the Object Data Modeling (ODM) tool, simplifying the process of defining schemas and performing database operations efficiently.

The platform uses Socket.IO to support real-time features such as booking updates, appointment confirmations, and notifications. Since no file-upload library or cloud storage system is used, all handyman information—such as certificates or documents—is currently managed manually or stored as simple text/data inside the database. This keeps the system lightweight and easy to maintain while still supporting essential service verification.

Key backend components include:

- **Runtime Environment:** The backend runs on **Node.js**, which provides a fast, event-driven environment ideal for handling multiple user requests efficiently.
- **Framework: Express.js** is used to build and manage the server routes, handle API requests, and implement authentication and middleware operations.
- **Database System:** The project uses **MongoDB**, a NoSQL database that stores all application data — including users, handymen, services, bookings, and reviews — in a flexible document format.
- **Data Modeling Library: Mongoose** is implemented to define schemas, enforce data structure, and simplify interaction with the MongoDB database.
- **Live Interaction: Socket.IO** enables real-time communication between customers and handymen, supporting instant appointment confirmations and live status updates.

5.1.3 Documentation Tools

For documentation, Microsoft Word was primarily used for writing the project report. Formatting and structure refinement were supported using standard document styling tools. Additionally, diagrams and system designs were created using:

- Draw.io: Class, Sequence, and Flow Diagrams
- Lucidchart / Figma: UI Mockups
- Canva / PowerPoint: Layout and Presentation Slides

These tools helped ensure clarity in visual representation and consistency in documentation formatting.

5.2 Testing

5.2.1 Unit Testing

In unit testing, each module of the FindMyVendor system was tested separately to ensure proper functionality before integration.

Table 1: Verifying Authentication (Login & Registration)

TC_ID	Test Case Description	Expected Output	Actual Output	Remarks
TC_01	Verify if a user can log in with invalid credentials.	Error message: *"Invalid email or password."*	Error message displayed.	Pass
TC_02	Verify if a user can log in with valid credentials.	Redirected to the customer/handy man dashboard.	User redirected successfully.	Pass
TC_03	Verify registration with an invalid email format.	Error message: *"Please enter a valid email address."*	Error message displayed.	Pass
TC_04	Verify password field with fewer than 8 characters.	Error message: *"Password must be at least 8 characters long."*	Error message displayed.	Pass

Table 2: Verifying Booking Functionality

TC_ID	Test Case Description	Expected Output	Actual Output	Remarks
TC_05	Create a booking with a valid customer and handyman.	Booking created successfully with status **"Pending."**	Booking created successfully.	Pass
TC_06	Cancel a booking before the handyman accepts it.	Booking status changes to **"Cancelled."**	Booking cancelled successfully.	Pass
TC_07	Handyman accepts the booking request.	Booking status changes to **"Accepted."**	Status updated successfully.	Pass

5.2.2 System Testing

System testing ensures the integrated system functions as intended across different user roles and modules.

Table 3: Verifying K-Handyman Discovery Page

TC_ID	Test Case Description	Expected Output	Actual Output	Remarks
TC_08	Search for nearby handymen based on user's current location.	List and map display of nearby handymen with distance information.	Handymen displayed correctly with accurate distance data.	Pass
TC_09	Apply service filter (e.g., plumbing or	Only handymen offering the selected service	Correctly filtered handymen shown based on	Pass

	electrical) during search.	are displayed.	selected service type.	

Table 4: Verifying Real-time Features

TC_ID	Test Case Description	Expected Output	Actual Output	Remarks
TC_10	Handyman updates the booking status (e.g., accepted, completed, or cancelled).	Real-time status change reflected instantly on the customer's dashboard.	Status updated instantly using Socket.io.	Pass
TC_11	Send chat message between customer and handyman.	Message appears instantly on both sides without page refresh.	Real-time chat working correctly.	Pass

Table 5: Verifying Profile Management

TC_ID	Test Case Description	Expected Output	Actual Output	Remarks
TC_12	Update customer or handyman profile information.	Updated details are saved and displayed correctly on the profile page.	Profile information updated successfully.	Pass
TC_13	Upload or change profile image for user or handyman.	Image uploaded and displayed correctly using the stored file path inside the system.	Profile image uploaded and displayed successfully.	Pass

5.3 Result Analysis

The K-Handyman web application successfully connects customers with skilled handymen through an efficient appointment-based system. The platform is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), providing a fast, secure, and user-friendly experience for both customers and service providers.

Smart Appointment Scheduling and Fair Allocation

The system integrates Bayesian Rating and Backtracking Algorithms to enhance service quality and efficiency. The Bayesian model ensures fair ranking of handymen based on their ratings and reviews, while the backtracking approach prevents overlapping appointments by checking availability and assigning optimal time slots automatically. This ensures that bookings are handled fairly and efficiently.Role-Based Access and User Management

User Roles and Access Management

The application supports two main roles — Customer and Handyman — each with dedicated dashboards and access privileges.

- Customers can register, log in, search for services, and book appointments.
- Handymen can view requests, accept or reject appointments, and manage their schedules. Role-based authentication using JWT tokens ensures secure access and data protection.

Real-Time Updates and Notifications

With Socket.IO, both customers and handymen receive instant notifications for booking confirmations, cancellations, and updates. This real-time communication improves transparency and keeps both parties informed throughout the service process.

Performance and Responsiveness

The system performs efficiently under multiple concurrent user interactions:

- Fast response time for location-based queries.
- Instant real-time updates during booking and chat operations.

- Smooth user experience with responsive React components.

Profile and Review System

Users and handymen can manage their profiles and view feedback from completed services. Profile photos or certificate images are stored and displayed using the system's local storage method instead of any cloud service. The review system helps customers choose reliable and trustworthy handymen.

Model and Algorithm Effectiveness

The integration of Bayesian rating and backtracking algorithms results in accurate and fair service matching. These methods ensure that highly rated handymen appear prominently while appointment scheduling remains conflict-free and optimized.

Chapter 6: CONCLUSION AND FUTURE IMPROVEMENTS

6.1 Conclusion

The K-Handyman project successfully provides a reliable and easy-to-use web platform that connects customers with skilled handymen for various home services. The system meets its primary goal of enabling users to search, book, and manage appointments efficiently through an integrated MERN stack architecture.

By implementing intelligent algorithms such as Bayesian Rating and Backtracking Scheduling, the system ensures fair handyman ranking and conflict-free appointment allocation. The inclusion of Socket.IO for real-time notifications enhances user interaction, providing instant updates for booking confirmations, status changes, and communication between customers and handymen.

Testing results confirm that the platform performs well under different conditions, including multiple concurrent users and booking operations. The system's performance, usability, and responsiveness align well with its intended objectives. Overall, K-Handyman demonstrates how modern web technologies can simplify home service management and create a fair, transparent, and efficient platform for both customers and service providers.

6.2 Future Improvements

- **Online Payment Integration:** Add payment gateways such as **Stripe**, **Khalti**, or **eSewa** for secure online transactions.
- **Mobile App Version:** Develop a mobile application using **React Native** to improve accessibility and convenience.
- **AI-Based Recommendation:** Enhance the handyman recommendation system using **machine learning** for personalized service suggestions.
- **Push Notifications:** Implement **Firebase Cloud Messaging (FCM)** for offline booking and update alerts.
- **Multilingual Support:** Introduce multiple language options to make the platform more inclusive and user-friendly.
- **Admin Dashboard:** Build an admin analytics dashboard to monitor system performance, track user activity, and review booking trends.

REFERENCES

[1] R. K. Thapa, *"Digital Platforms for Local Service Management: A Case of South Asian Urban Innovation," Journal of Emerging Web Technologies*, vol. 6, no. 2, pp. 25–38, 2022.

[2] JWT.io, *"Understanding JSON Web Tokens (JWT),"* [Online]. Available: https://jwt.io/introduction. [Accessed: Jul. 4, 2025].

[3] MongoDB Documentation, *"MERN Stack Overview: Building Full-Stack JavaScript Applications,"* [Online]. Available: https://www.mongodb.com/mern-stack. [Accessed: Jul. 4, 2025].

[4] Socket.IO Team, *"Real-Time Communication for Web Applications,"* [Online]. Available: https://socket.io/docs/v4. [Accessed: Jul. 4, 2025].

[5] React.js Foundation, *"React: A JavaScript Library for Building User Interfaces,"* [Online]. Available: https://react.dev. [Accessed: Jul. 4, 2025].

[6] Node.js Foundation, *"Node.js Developer Guide,"* [Online]. Available: https://nodejs.org/en/docs. [Accessed: Jul. 4, 2025].

APPENDIX

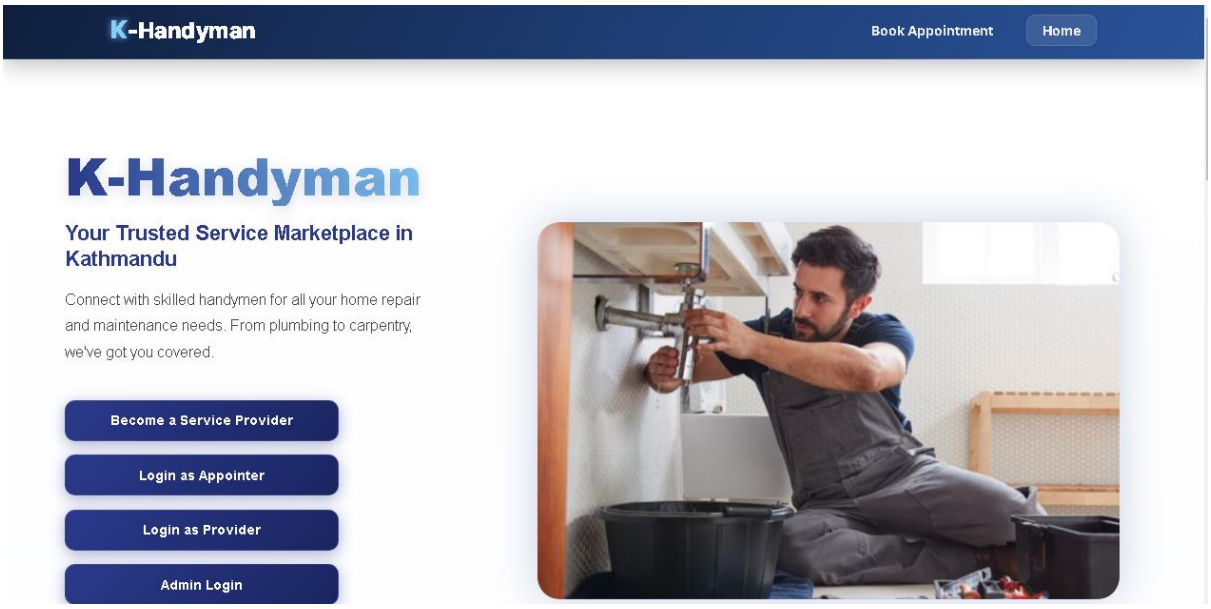


Figure:8 Home Page

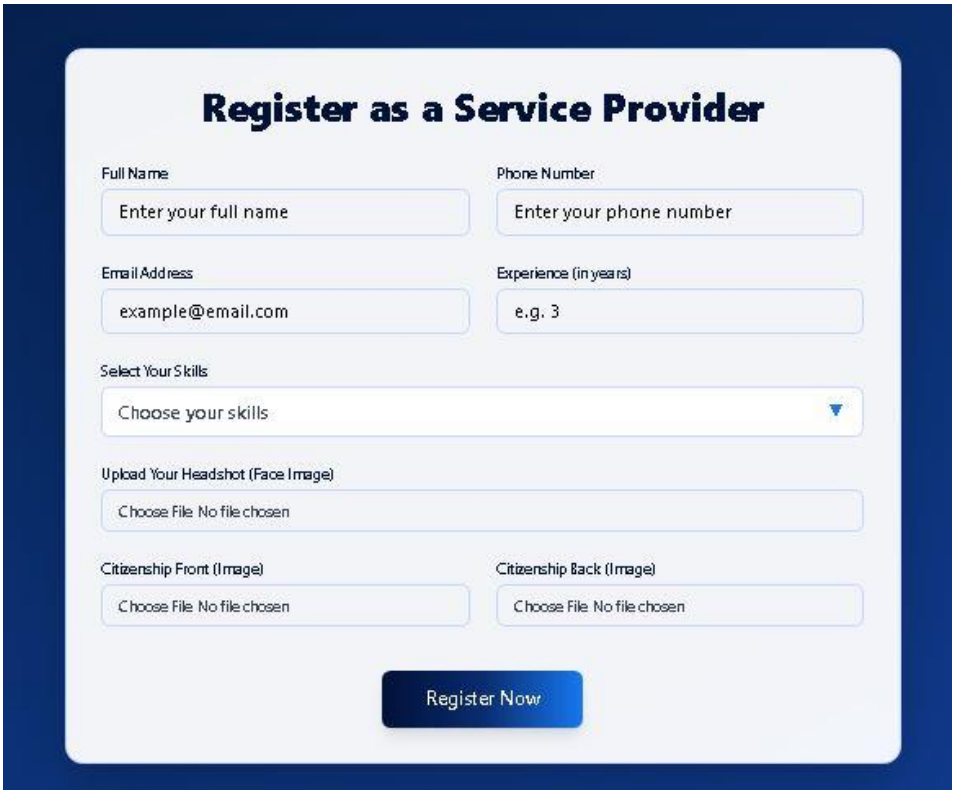


Figure :9 Register Page

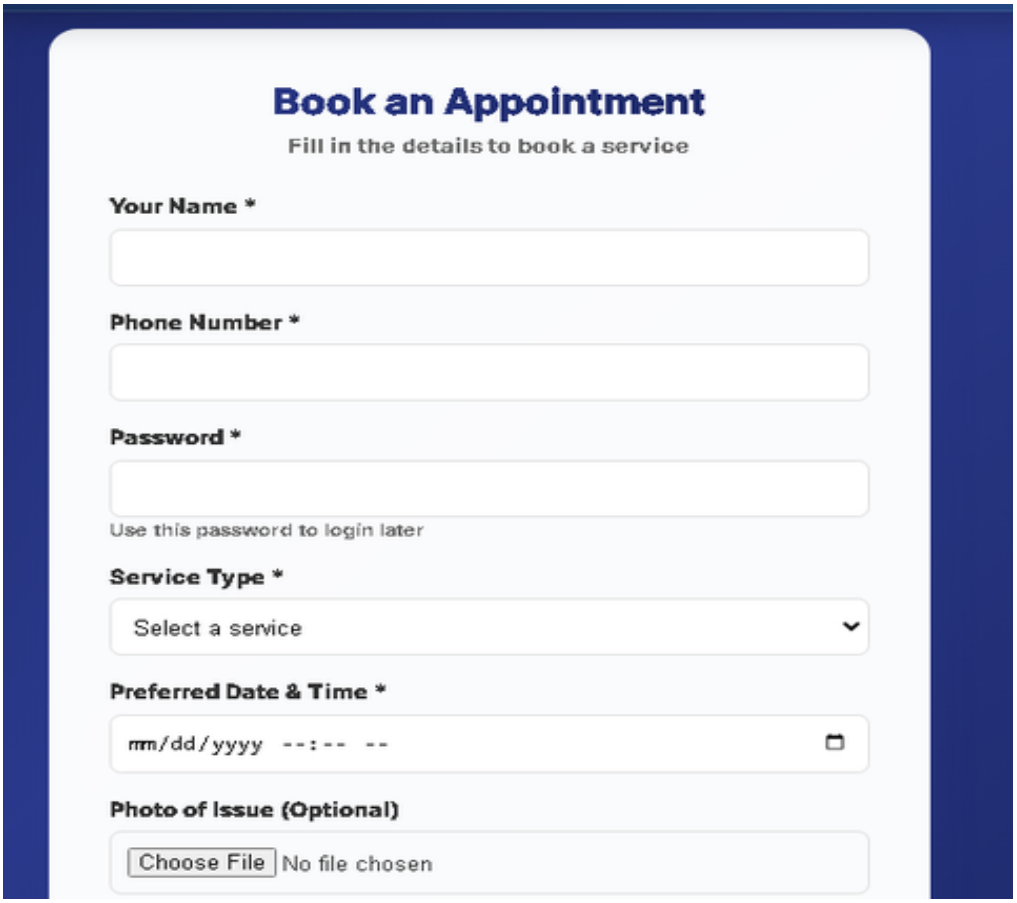


Figure:10 Login Page/Booking Page

Figure:11 Discover Page/Service Page

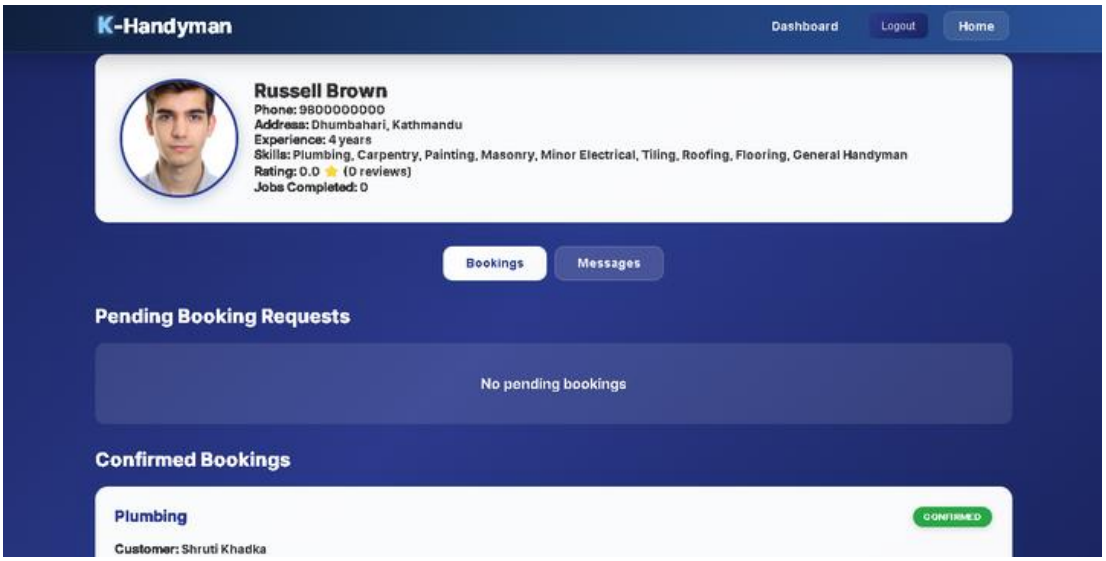


Figure:12 Service Provider Page

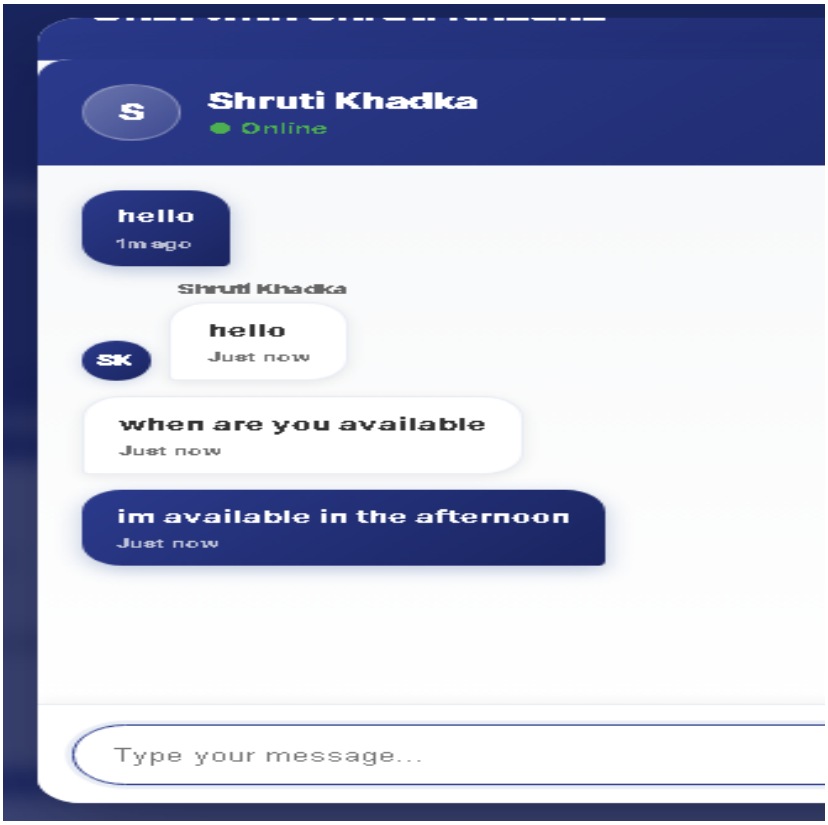


Figure:13 Chat between Appointer and Service Provider

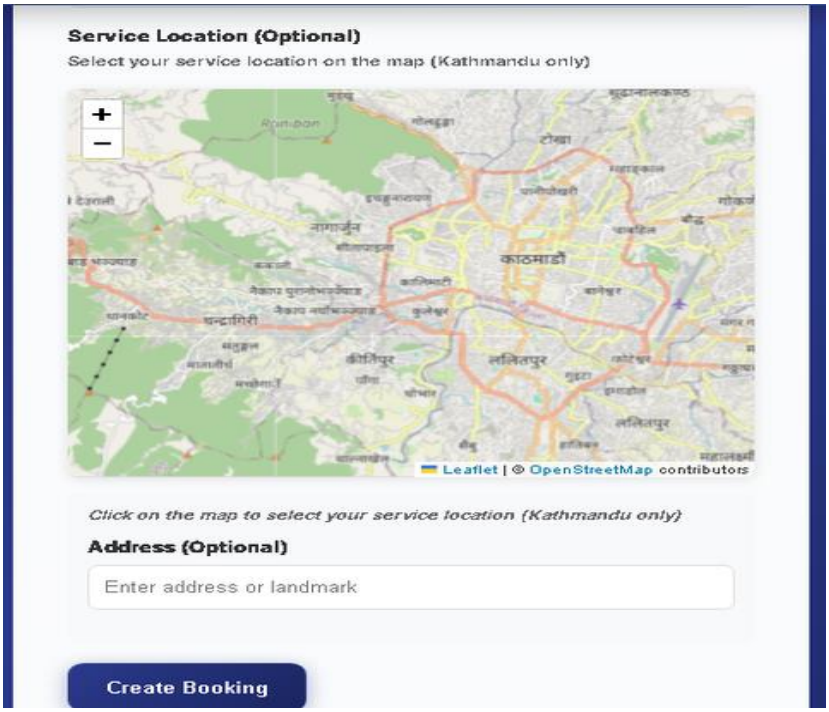
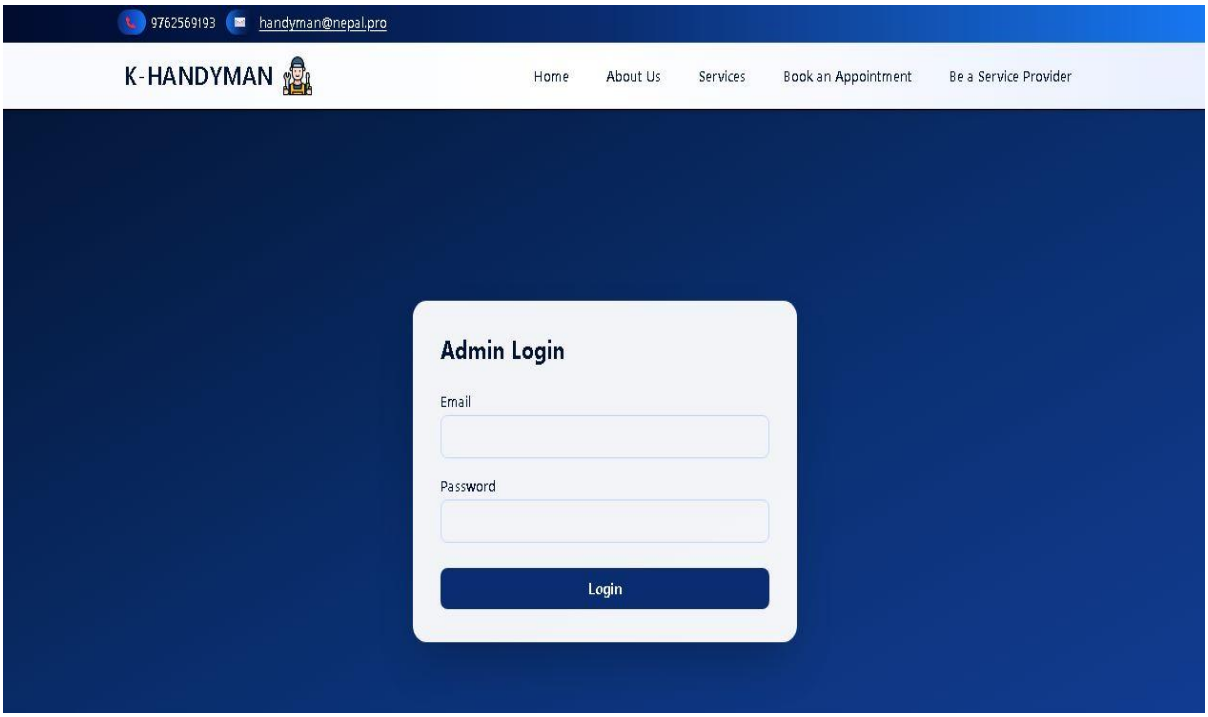


Figure:14 Service Location Page



Fiure:15 Admin Login Page

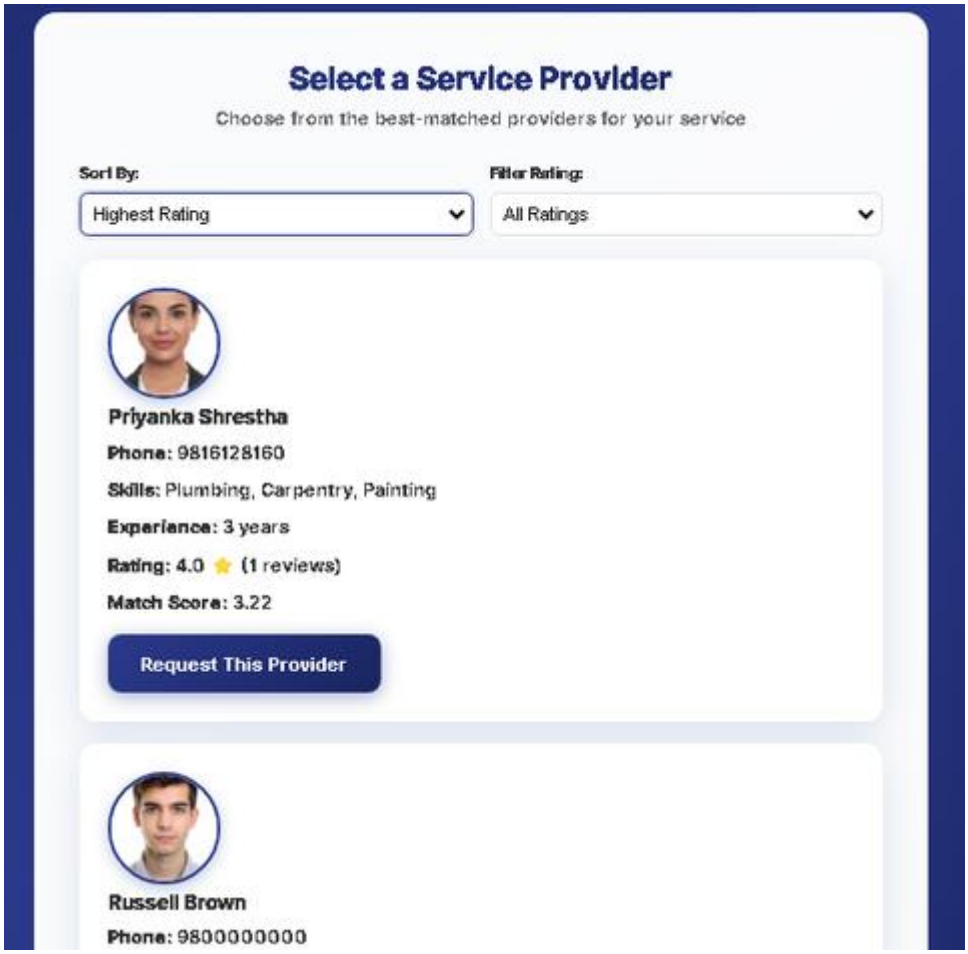


Figure:16 Algorithm Based Service Provider Ranking